

Banking Stress Test Handbook

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Abstract

After the Global Financial Crisis of 2007–2009, stress testing has emerged as a critical tool in prudential supervision and regulation. This report reviews current stress testing practices and critically assesses their benefits and shortcomings. In addition, it outlines a guideline to implement a banking stress test framework using publicly available data. Our guideline covers data requirements and sources, the data collection process, scenario building, risk modelling methodologies, and a balance sheet transmission process. The guideline also demonstrates analysis and policy implications that can be inferred from stress testing results.

JEL classification: G21, E58, G28

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Abbreviations

AFS	available for sale
BICS	Bloomberg Industry Classification Standard
BQL	Bloomberg Query Language
CAR	capital adequacy ratio
FVTPL	fair value through profit and loss
FVOCI	fair value through other comprehensive income
GDP	gross domestic product
IDR	Indonsian rupiah
IFRS	International Financial Reporting Standards
JCI	Jakarta Composite Index
LLP	loan loss provisions
MTM	mark to market
NPL	non-performing loans

1. Introduction

A banking stress test is an analysis conducted by either financial supervisors or banks to examine whether the banking sector has enough capital to withstand a negative economic shock under hypothetical scenarios. From the perspective of central banks and bank regulators, stress tests have the potential to support both risk measurement and risk management (Bank of England, 2016). For example, stress-test results help policymakers set capital requirements, which are in place to ensure banks fund themselves with sufficient loss-absorbing capital to reduce their likelihood of failure.

The now-widespread practice of stress testing has been carried out by authorities since the early 2000s, mainly as part of the International Monetary Fund's Financial Sector Assessment Program. However, both policymakers and academic experts argue that the bank stress testing framework before the 2007–2009 financial crisis was not sufficient to accurately depict conditions related to financial stability and could not produce enough information to bring about an adequate policy response (Buncic & Melecky 2013; Čihák 2007; Galati & Moessner 2013). After the crisis, many policymakers realised the importance of prudential supervision and the urgency to establish a crisis management protocol. As a consequence, banking stress testing has become one of the most prominent tools to assess financial stability amongst prudential authorities (Acharya et al. 2014; Hanson et al. 2011; Hirtle et al. 2009). Lessons from the crisis have led many central banks and financial authorities to start developing new generations of stress tests with more sophisticated designs (Foglia 2009).

The development of new stress test generations has improved almost every aspect of the design of bank stress tests (Cerutti & Schmieder 2014). For instance, scenario construction has developed from a basic sensitivity analysis to a more complex, model-based scenario (Basel Committee on Banking Supervision 2009). In addition, the complexity and sophistication of the econometric models have been markedly refined for better estimating possible losses under stress scenarios, particularly in terms of credit risk modelling (see Buncic & Melecky 2013; Ishikawa et al. 2012; Schechtman & Gaglianone 2012). Despite the remarkable methodological improvements of recent years, some challenges and weaknesses remain to be tackled (Borio et al. 2014; Cerutti & Schmieder 2014; Galati & Moessner 2013).

Against this background, this report aims to provide guidelines for conducting banking stress tests. Chapter 2 of this report presents the conceptual framework, including risk identification, scenario construction, and risk modelling. Chapter 3 describes the data and methodology. Chapter 4 presents the results and interpretations. Chapter 5 concludes the report and outlines potential future improvements.

2. Conceptual framework

Risk identification and scenario construction

The construction of macroeconomic scenarios is usually the first step in executing a stress test. This step, however, must be preceded by a determination of risk appetite and an identification of risk factors. Risk appetite usually influences the severity and the length of the stress test scenarios (Schuermann 2014). Meanwhile, the identified risk factors usually affect the number of variables used in the scenarios. As more risks are identified, the scenarios will become more complex.

For the construction of macroeconomic scenarios, the earliest stress test designs usually applied a sensitivity approach. This approach usually shocks one variable based on the desired risk appetite and estimates the impact on the banking sector while holding other factors constant. Yet despite its straightforward way to deliver a quick assessment of financial system vulnerability in responding to certain risk factors, this approach has several weaknesses. First, the financial system is heavily interconnected, hence the movement of a variable is very likely to affect other macro-financial variables. As a consequence, it is impossible to isolate the impact of one particular shock while neglecting its interaction with other variables in the financial system. Second, given its objective as a prudential tool, one of the features of a banking stress test is to include a severe but plausible scenario. Under financial distress, several financial indicators may deteriorate simultaneously, leading to financial vulnerabilities and imbalances. Thus, shocking only one variable while holding other variables constant may lead to counter-intuitive results.

Accordingly, the creation of scenarios which are more comprehensive is preferable to a basic sensitivity analysis for capturing the interactions between financial and macroeconomic variables. Several risk factors should be shocked simultaneously to capture these interactions. In most cases, such simultaneous shocks are constructed based on historical events – the rationale being that events which have happened in the past may happen again in the future. These scenarios suit the principle of 'severe but plausible' since they integrate risks that have materialised previously. Nevertheless, this approach is not restrictive (i.e. can be subject to some degree of flexibility) for several reasons. First, in many cases historical scenarios are constructed based on the most recent financial crisis, which means that the stress test only aims at replicating that crisis and isolates other possible risks from the scenario. The credibility of the stress test may therefore wane if the most recent crisis occurred a long time ago, particularly if the economy has since gone through a structural adjustment that diminishes the relevance of such a scenario. Second, due to the rapidly changing economic condition, historical scenarios may underestimate the evolving risk levels and interactions (Basel Committee on Banking Supervision 2009). Due to these limitations, hypothetical scenarios can be constructed to complement historical scenarios with regard to capturing risks or events that have never been experienced before. This approach can be justified since plausible events do not necessarily have to be events that have happened in the past, but can also be those predicted to happen in the future. In taking this approach, stress test design will retain its forward-looking characteristics and be capable of adapting to transformations in the financial system.

Figure 1. Illustration of a brief banking stress test methodology



Banking stress test starts with macro-scenario modelling. These hypothetical macroeconomic scenarios are used to forecast future non-performing loans (NPLs) and loan loss provisions (LLPs). Increments in LLPs will affect key financial metrics such as earnings after tax (EAT) and capital adequacy ratio (CAR).

Credit risk modelling

After identifying the risk factors and constructing the stress testing scenarios, another important step in banking stress testing is to model the risk transmission. Such modelling is crucial since it provides a linkage between the macroeconomic scenarios and risk factors in the banking sector. The stressed risk factors will affect the monitored financial indicators such as capital adequacy, non-performing loans, profitability etc. The literature suggests two main approaches to translating the impact of macroeconomic variables on financial soundness indicators: the balance sheet approach and the integrated approach (Sorge & Virolainen 2006; van den End & Tabbae, 2006). The former mainly uses the estimated coefficients of macroeconomic variables from econometric models as a simulation basis to find the impact of these stressed variables on the financial system. The model can be either structural or 'reduced form' using time series or panel data techniques and might only consider one risk factor at a time. Meanwhile, the latter follows a more complex technique – taking an integrated approach which combines multiple risk factors (particularly credit and market risks), putting them into a single model to estimate the increased probability of losses under stress scenarios.

One aspect that needs to be considered in risk modelling is the assumption about linkage behaviour between macroeconomic scenarios and risk factors in the financial system. Previous studies have shown that the sensitivity of risk factors in the financial system tends to be higher in stressed conditions (crisis periods) compared with normal times (Bangia et al. 2002; Virolainen 2004). This indicates there are dynamics in financial systems under duress which cause the linkage between macroeconomic scenarios and risk factors to be non-linear. In this regard, Arnold et al. (2012) argue that some of the earlier stress test designs failed to capture these dynamics. Consequentially, these models may under-estimate the risk factors embedded in the financial system. Therefore, regardless of the linkage approaches used by stress test modellers, a stress test model must be able to capture these dynamics to address the non-linearities.

From the modelling point-of-view, this can be captured by non-linear models, which some more recent stress test designs have attempted. For instance, regime-switching models have been used as a possible option (Borio et al. 2014). Juselius and Kim (2011) have shown that this approach can capture the elasticity of bank variables in responding to macroeconomic shocks, depending on state-varying conditions. Another possible approach is by estimating quantile regression models (for empirical examples see Covas 2014; Ishikawa et al. 2012; Surjaningsih 2014; Schechtman & Gaglianone 2012). This approach can estimate financial distress based on the selected quantile that suits the constructed scenario. Some stress test designs also employ a probabilistic approach by focusing on the tail of conditional distribution to represent the stress scenario (see Schechtman & Gaglianone, 2012). In some cases, systemic risk measurements such as CoVaR and marginal expected shortfall can be used as the basis for financial indicators in estimating the conditional tail (Acharya et al. 2014; Puzanova & Düllmann 2013; Schechtman & Gaglianone 2012).

Apart from the modelling technique, the selection of variables in the model also matters. In most of the existing stress test designs, standard macroeconomic variables such as GDP growth, CPI, unemployment rate, exchange rate fluctuation and interest rate are included in the model (Buncic & Melecky 2013; Schechtman & Gaglianone 2012; Vazquez et al. 2012). However, this feature is not restrictive. In some cases, additional variables can be added such as the property prices index, automotive sales, foreign loans, commodity prices, household expenditures etc. (Ganbaatar & Selenge 2012; Ishikawa et al. 2012). This flexibility is allowed since a stress test model needs to reflect the behaviour of the exercised financial system, thus providing a relatively more accurate result. More advanced econometric models and appropriate variable selection will, of course, create a more sophisticated stress test design of the kind usually preferred by prudential supervisors and stress test users. Nevertheless, a more sophisticated design depends heavily on supporting infrastructure, particularly data availability and granularity level. In its simplest form, prudential top-down stress testing uses aggregate financial data and examines the impact of a stress scenario on the financial system as a whole. For example, prudential supervisors can simulate the impact of a stress scenario on the industry capital adequacy ratio, non-performing loans and profitability. Despite its simplicity, this technique does not capture idiosyncratic information that relates to financial institutions in the system. Hence, while prudential supervisors are able to assess the resilience level of the financial system, they cannot investigate which financial institutions are affected the most by the stress scenario.

Due to this limitation, most of the existing stress test designs employ institutional-level data instead of aggregate data since these can provide more fruitful analyses. The use of institutional-level data makes it possible to examine which institutions are most vulnerable under a particular stress scenario and, to some extent, can help reveal the most influential risk factors for each financial institution. When data infrastructure is available, institutional-level data can be broken down even further to an unconsolidated level, which treats the parent and its subsidiaries as different entities, thus improving stress testing sensitivity. This level of granularity can capture cross-banking capital and liquidity relocation, which becomes particularly relevant for large cross-border banks (Cerutti & Schmieder 2014).

3. Data sources and collection

Macroeconomic data

Macroeconomic-level data are utilised to represent macroeconomic factors that affect the balance sheets and income statements of banks. Under this guideline, the framework employs four macroeconomic variables that are significant and consistent in explaining the effect of changes in credit risk and market risk on banking profitability. Each of the GDP growth, inflation, property price, and the repo rate variables explain changes in non-performing loans. On the other hand, changes in bond prices, which are also derived from the repo rate, measure the source of banking losses arising from available-for-sale and marked-to-market trading securities. All macroeconomic data employed in the stress testing exercise in this chapter are released publicly on a quarterly basis. Table 1 presents the variables along with their definitions and sources.

Table 1. Macroeconomic variables

Variable	Definition	Source
gdp	GDP growth YoY	BPS
d_property	Changes in property index QoQ	QoQ changes of 18 cities' residential property index from SHPR BI
reporate	BI rate	BI rate
d_price	Changes in bond price QoQ	Calculation based on BI rate, i.e. $d_price = -10 * (BI\ rate_Q - BI\ rate_{Q-1})$

Bank-level data

The usage of bank-level data is intended to capture how the proposed macroeconomic scenarios affect the balance sheets and income statements of banks. Using individual bank-level data instead of industry aggregates allows for a more granular and thorough assessment of how each bank would be affected by shocks, as well as the identification of banks with the highest risk of collapsing. This stress testing exercise utilises data from 48 commercial banks that are publicly listed on the Indonesia Stock Exchange. This sample is representative of Indonesia's banking industry as the total assets of the banks within the sample make up approximately 85% of the banking industry's assets. Bank-level variables consist of financial accounts from the banks' balance sheets and income statements, as well as some key financial ratios such as the tier 1 capital ratio and the total risk-based capital ratio, all of which are publicly available through each individual bank's financial statements and catalogued by Bloomberg. The complete list of the bank-level variables is summarised in Table 2.

Data collection using Bloomberg Query Language

Due to the time-consuming nature of collecting data from individual banks for an industry-wide stress test, the use of third-party software to automate data collection and cleaning is highly encouraged. For this stress testing exercise, data collection is assisted by the Bloomberg Query Language (BQL) through software known as Bloomberg Terminal, to which most financial institutions and regulators around the world subscribe. The BQL allows users to access Bloomberg's

financial data from various securities and companies through specific queries in Microsoft Excel, which facilitates easier and quicker data transformation and analysis. Querying the data in Excel also allows users to efficiently import the data to Stata, the software that this exercise uses for stress testing calculations. This approach minimises the possibility of human error being introduced to the data collection process.

Table 2. Bank-level variables

Variable	Definition	Source
BS_TOT_ASSET	Total assets	Bloomberg
BS_TOT_LOAN	Total loans	Bloomberg
BS_NON_PERFORM_LOANS	Non-performing loans	Bloomberg
BS_CUSTOMER_DEPOSITS	Total deposits	Bloomberg
BS_RISK_WEIGHTED_ASSETS	Risk-weighted assets	Bloomberg
BS_TIER1_CAPITAL	Tier 1 capital	Bloomberg
BS_TOT_CAP_FUND	Total capital funds	Bloomberg
BS_TIER1_CAP_RATIO	Tier 1 capital ratio (%)	Bloomberg
BS_TOT_CAP_TO_RISK_BASE_CAP	Total risk-based capital ratio (%)	Bloomberg
IS_INT_INC	Interest income	Bloomberg
IS_PROV_FOR_LOAN_LOSS	Provision for loan losses	Bloomberg
NON_INT_INC	Total non-interest income	Bloomberg
NON_INT_EXP	Total non-interest expense	Bloomberg
NET_INT_INC	Net interest income	Bloomberg
IS_CURRENT_INCOME_TAX_BENEFIT	Income tax expenses – current	Bloomberg
IS_DEFERRED_INCOME_TAX_BENEFIT	Income tax expenses – deferred	Bloomberg
IS_RSRV_CHRG	Zakat expenses	Bloomberg
NET_INCOME	Net income	Bloomberg
TRADING_SECS_FA_AT_FV_THRU_PL	Trading securities and financial assets at fair value through profit and loss (FVTPL)	Bloomberg
AV_SA_FA_AT_FV_THRU_OT_COMP_INC	Available for sales financial assets at fair value through other comprehensive income (FVOCI)	Bloomberg

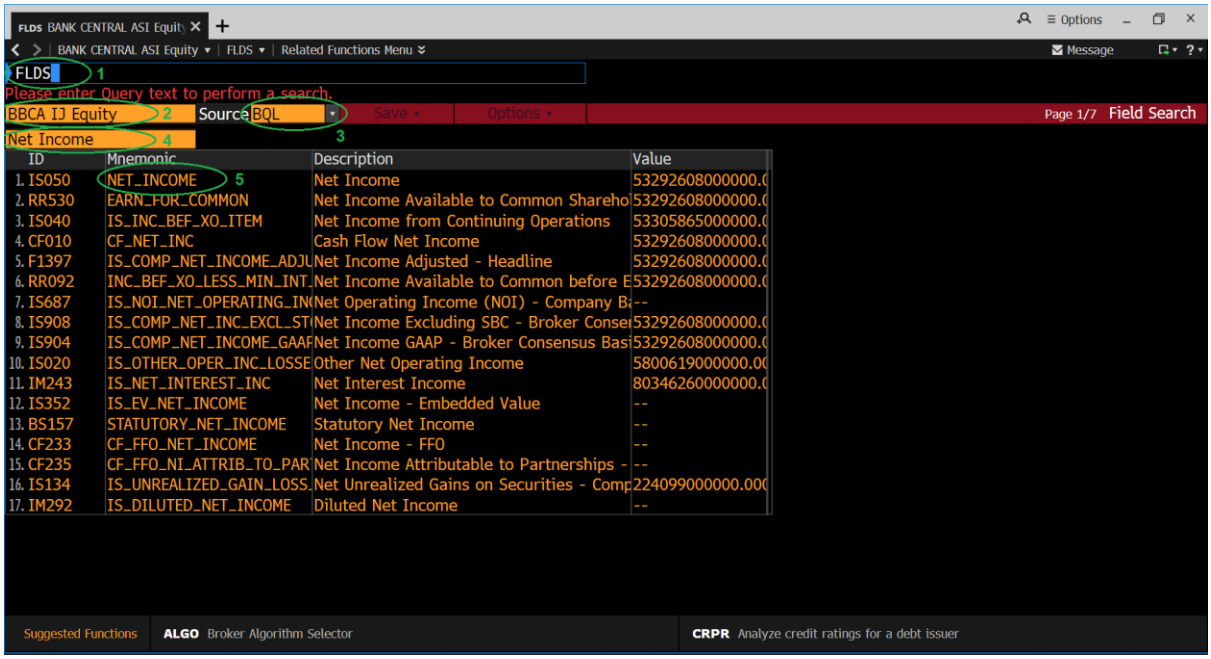
To make a data query using BQL, users need to first understand how Bloomberg stores and catalogues its data. Two recurring concepts in the BQL are "universes" and "data items". A universe is a set of one or more Bloomberg securities whose data will be collected. For example, a universe of "BBCA IJ Equity" will result in data being pulled only from the company PT Bank Central Asia Tbk, while a universe consisting of "JCI Index" will result in data being pulled from the Jakarta Composite Index (JCI). Filtering rules can also be used to simultaneously pull data from multiple securities; for example, it is possible to extract data from every single publicly listed company in the Indonesia Stock Exchange by setting the universe to members of "JCI Index". The technical details underpinning the filtering rules will be explained later.

Data items are specific types of data or observations that need to be extracted from a certain security. A single Bloomberg security contains various data items corresponding to the security,

each with a unique ticker; for example, the security "BBCA IJ Equity" contains key financial metrics of the company such as the shares outstanding (BS_SH_OUT), total assets (BS_TOT_ASSET), net income (NET_INCOME) etc. Since not all securities contain the same data items, it is important for users to identify which data items to extract from the security, which the Bloomberg Terminal's Data Field Finder can assist with. To use this Bloomberg function, users can follow the step-by-step instructions and illustration (Figure 2) below:

1. Access the Data Field Finder by entering "FLDS" into Bloomberg's search bar and choosing the function
2. Enter the name of the security in the top-left Securities box
3. Click the drop-down Source box and choose the "BQL" option
4. Type the desired data item in the Enter Query box
5. Scroll down the list for the appropriate data item and use the ticker in the Mnemonic column

Figure 2. Illustration of Bloomberg Data Field Finder (FLDS)

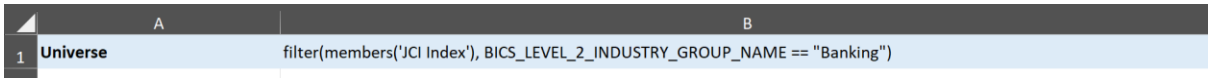


After determining the desired universe and data items, users can start building their query with the BQL formula in Microsoft Excel. The BQL formula syntax can simply be expressed as:

```
=BQL("Universe"; "Data Item")
```

To make the formula easier to adjust, the universe and data items do not have to be explicitly stated in the formula but can be replaced with Excel cell references. This makes changing the universe and/or data items easier since the tickers are placed in separate cells. For this exercise, due to the large amount of data items being pulled, the universe and data items are put in a different sheet to the data output. The Excel template provided with this stress testing guideline consists of two sheets: "Dictionary", which is the list of universes and data items, and "Output", which is the resulting data.

Figure 3. Bloomberg BQL Universe in Microsoft Excel



The first row of the "Dictionary" sheet contains the universe of the query. To capture the desired bank-level data in the universe, filtering rules are applied through the BQL **filter()** function to select securities that match more than one criteria or condition. As shown in Figure 3, the universe filtering rules in the Excel template consist of two conditions: the first is members of the JCI Index, corresponding to all publicly listed companies in the Indonesia Stock Exchange; and the second is companies classified as banking industry based on the Bloomberg Industry Classification Standard (BICS). The resulting universe is the set of companies classified as banking industry **and** publicly listed in the Indonesia Stock Exchange, thus matching the desired criteria.

Figure 4. Bloomberg BQL data item construction in Microsoft Excel

Field	Range	Unit
BS_TOT_ASSET	(fill=prev, dates=range(start=-10y, end=0d, frq=q)).value	*0.000001
BS_TOT_LOAN	(fill=prev, dates=range(start=-10y, end=0d, frq=q)).value	*0.000001

The third row onwards contain the data items, which are constructed from components separated into three different columns: Field, Range, and Unit. Figure 4 shows that the Field column contains the specific data item ticker that needs to be extracted; for example, BS_TOT_ASSET refers to the company's total assets. The Range column contains the observation range of the data, which can be broken down to three parts: **fill=prev** automatically fills in the latest previous observation if the data are unavailable at a given observation, **dates=range()** specifies the observation range and frequency for the data, which in this example is 10 years prior in quarterly frequency, and **.value** pulls the numerical value of the data field. Finally, the Unit column scales down the value of the observation; in this case, a multiplication of 0.000001 is used for variables in nominal (IDR) form to convert the unit into millions, while the column is left empty for variables in ratio form.

The complete list of data items corresponds to the list of bank-level variables given in the previous section, with one exception: the data item of shares outstanding (BS_SH_OUT) is used simply as a placeholder for the observation date. Information from the three columns are then combined into one data item with Excel's **=CONCATENATE()** formula, resulting in the Item column. More information on how to manipulate the data fields and range can be found in the Bloomberg BQL Help Page, but changes beyond the observation range start and end dates should not be necessary for this stress testing exercise.

With the universe and the complete set of data items defined in the Dictionary sheet, making the data query in the Output sheet can be done by simply referring to the tickers in the Dictionary sheet. Thus, the BQL formula for the query can be written as:

```
=BQL(Dictionary!B1; Dictionary!B4:B24)
```

The cell where this formula is entered becomes the top-leftmost cell of the resulting data table, so users should enter the formula in cell A1 of the Output sheet in the Excel file. The resulting data will be in a long panel format where each row/observation represents a bank at a given year and each column/variable represents a data item. However, due to the limitations of BQL, observations in some quarters will be repeated if Bloomberg does not have information on a given bank's financial statements for a certain period. To address this issue, additional cleaning of the data is required, which can be conducted using Stata (explained in the next section).

4. Methodology and framework

Scenario building

This guideline uses three scenarios, each representing different macroeconomic situations. The first scenario is the baseline scenario (Scenario 1), where there is no significant disruption in the economy and each of our five macroeconomic variables moves in the way that an ordinary forecast would predict. Scenario 1 is the baseline for this stress test exercise, which means it is our basis for comparison with the effects of the shocks that will be employed in the next two scenarios.

Scenario 2 and Scenario 3 are our adverse scenarios, where the shocks are haircuts to each of our five macroeconomic variables. The haircut is based on the standard deviation obtained from 10-year quarterly data for each of the macroeconomic variables. The shocks are linear in the sense that for each predicted future quarter, the same magnitude of haircut will be employed to the five variables.

For Scenario 2, this stress test uses a 0.25 standard deviation haircut per quarter. For Scenario 3, the haircut is a 0.5 standard deviation per quarter. These haircuts will be a subtraction to the GDP growth, bond price and property price change, but an addition to the repo rate and inflation. These subtractions and additions will occur in each predicted quarter with the same magnitude for each variable, according to the scenarios being employed.

Table 3. Scenario (linear) building process

Quarter	GDP (gdp)			BI Rate (reporate)			Housing Prices (property)			Bond Prices (d_price)		
	S1	S2	S3	S1	S2	S3	S1	S2	S3	S1	S2	S3
2024Q1	4,94	4,94	4,94	6,00	6,00	6,00	120,93	120,93	120,93	-2,50	-2,50	-2,50
2024Q2	4,84	4,25	3,66	6,25	6,48	6,70	121,93	118,12	114,31	-2,50	-3,51	-4,51
2024Q3	4,83	3,66	2,49	6,25	6,70	7,16	121,93	114,31	106,69	-2,50	-4,51	-6,52
2024Q4	4,79	3,08	1,31	6,25	6,93	7,61	121,93	110,50	99,07	-2,50	-5,52	-8,53
2025Q1	4,79	2,49	0,14	6,25	7,16	8,06	121,93	106,69	91,44	-2,50	-6,52	-10,5

S1 refers to Scenario 1, the baseline, which reflects ordinary forecasting. S2 is Scenario 2, where a shock of 0.25 standard deviation hits each macroeconomic indicator in each quarter. S3 is Scenario 3, which has a similar construction but with 0.5 standard deviation. These specific linear macro scenarios are an example from the 2024 Q1 surveillance, when there was a belief of an ongoing BI rate hike trend.

There is room to modify these scenarios by considering changes in general economic trends. For example, when BI repo rate cuts are trending, then the baseline (Scenario 1) would forecast a rate cut in future quarters. It would then be sensible to apply a 0.25 standard deviation to Scenario 2 (or less), subtracting to the repo rate rather than adding. Thus, the shock scenarios can be recalibrated according to the prevailing economic and policy trends. The user's expert judgment is required to properly identify realistic as well as significant adverse scenarios and doing so opens

up the possibility of employing a non-linear shock. The table below illustrates the scenario generation process with constant shocks. S1 denotes Scenario 1 and so on.

Preliminary preparations and data processing

This stage starts the calculation procedure of banking stress testing using Stata. For convenience in accessing the folder path, it is important to set up global paths. These global paths determine the four main folders that the user will access most frequently. As illustrated in the do file (a file in Stata that records the command that needs to be run) sheet below, this stress test identified four key paths: 1) "source" – the path to obtain the Excel sheet that contains the macroeconomic data and scenarios; 2) "bql" – directly connects to the folder in the Bloomberg terminal where the banking data Excel sheet is being stored; 3) "data" – the folder where the previously mentioned data will be stored or re-imported; 4) and "result" – the folder where the result of the test will be stored. Users can modify these global path arrangements to adjust to their preferences (by changing the directory paths to suit the file locations).

Figure 5. Global paths determination using Stata

```
1  ****************************
2  ****** Global Settings ******
3  ****************************
4
5  *   Notes: The first phase is to determine global paths. We define the path for data source, Stata storage, and results storage.
6
7      global source   "Z:\Shared\PROSPERA\Finance\BT.2.2_Finance Policy Team\00. Activities\11. Team Activities\g. Data Master File"
8                          // Select path for macro raw data source
9
10     global bql "Z:\Shared\PROSPERA\Bloomberg\Finance"
11                         // Select path for bank raw data source from bloomberg terminal
12
13     global data     "Z:\Shared\PROSPERA\Finance\BT.2.2_Finance Policy Team\00. Activities\06. KSSK Meeting\a. Banking Surveillance and Stress
   Test\2024Q2\Stata"                          // Select path for Stata data results
14
15     global results  "Z:\Shared\PROSPERA\Finance\BT.2.2_Finance Policy Team\00. Activities\06. KSSK Meeting\a. Banking Surveillance and Stress
   Test\2024Q2\Result"                         // Select path for stress test results
16
17     *ssc install lgraph, replace
18
```

In line 17 in the do file above, there is a line that commands Stata to install an additional package. It is important to install this package as it is the command that supports the creation of graphs and charts. Note that it only needs to be installed once. For users who have not yet installed the package, please delete the asterisk "*" in front of the **"ssc install…"** and run the line 17 command. Once the installation is complete, feel free to put the asterisk back in to avoid the command being run numerous times when conducting future stress testing.

The next procedure is to save the macroeconomic data and scenarios that have been constructed. As illustrated in the do file below (Figure 6), in line 26, **"$source…"** is the global path that has been set up previously. It instructs Stata to know where to find the folder that hosts the macroeconomic data Excel sheet. This line 26 instructs Stata to import the Excel file named "Solvency ST", look for the sheet "Macro" within the Excel file, and then set the first row in that sheet as the variable name. Line 27 applies the format command of **"%tq"** after the variable "quarter" to notify Stata that the time dimension is quarterly. Line 28 saves the data into a Stata data format "macro.dta".

Starting from line 31, using a looping mechanism, Stata then collects the sheet consisting of Scenario 1 to 3 from the Solvency ST and proceeds to save each of the sheets into a separate dta file. Each of these separate scenario datasets will then be appended to the previously stored "macro.dta", after which they are stored in separate dta files. Each of these dta files contains the "macro.dta" that has been combined with different scenarios (comb 1.dta is the appended macro.dta with scen 1.dta, and similarly comb 2.dta and comb 3.dta).

Figure 6. Importing macroeconomic data and scenarios using Stata

```
18
19  ****************************
20  **** Raw Data Extraction ****
21  ****************************
22
23  *   Notes: The second phase is to import macroeconomic data, macroeconomic scenarios, and bank-level data from excel raw data into Stata format (dta).
24
25  *1. Macroeconomic data
26      import excel "$source\Solvency ST.xlsx", sheet("Macro") firstrow case(lower) clear                 // Import macroeconomic data
27      format quarter %tq
28      save "$data\macro.dta", replace
29
30  *2. Macroeconomic scenario
31      forval i=1/3{
32          import excel "$source\Solvency ST.xlsx", sheet("Scenario `i'") case(lower) firstrow clear       // Import stress test macro scenario
33          save "$data\scen`i'.dta", replace
34          use "$data\macro.dta", clear
35          append using "$data\scen`i'.dta"
36          gen ln_forex=log(forex)
37          save "$data\comb`i'.dta", replace
38      }
39
```

After saving the macroeconomic scenarios into different datasets, this exercise then proceeds to the importation and cleaning of bank-level data, which has been collected using the BQL commands. This entire process is illustrated in Figure 7. First, after importing the Output sheet of the BQL Excel file, the variables are renamed using the "*" wildcard to remove the remaining BQL range instructions. After that, the variable "bs_sh_out" is redesignated as the date variable, and a new variable "quarter" is generated to signify the observation date in quarterly format.

Figure 7. Importing BQL data from Excel to Stata

```
40  *3. Bank-level data
41      import excel "$bql\BQL Test.xlsx", sheet("Output Sheet") firstrow case(lower) clear
42      rename *fill* *
43      rename is_*_benefit* is_*_benefit
44
45      rename bs_sh_out dates
46      gen quarter = qofd(dates)
47      format quarter %tq
48      drop if quarter == .
49
50      rename a id
51      gen bank = substr(id, 1, 4)
52      bys bank quarter: gen dup = cond(_N==1, 0, _n)
53      drop if dup > 1
54      isid bank quarter
55      drop dup
56
57      merge m:m bank using "$data\kbmi.dta"
58      drop if _merge == 1
59      drop _merge
60
61      encode bank, gen(bank_id)
62      xtset bank_id quarter
63
64      drop if bank_id == 7 //temporary command, to exclude outlier
65
```

Next, the variable "bank" is generated by extracting the first four letters of the security name using Stata's **"substr()"** function such as in line 51. In this case, "substr(id, 1, 4)" instructs the program to get a substring from the variable "id" starting at character 1 with a length of 4 characters, thus yielding the four-letter bank stock ticker. After that, the command **"bys"** (short for bysort) is used to sort the observations by bank ticker and quarter, and then create the variable **"dup"** to tag duplicate observations. These duplicates are then dropped, after a check that the data is uniquely identified by bank tickers and quarters. This process is illustrated in lines 52 to 55.

To incorporate information on the bank's classification based on capital (KBMI), the dataset is then merged with the "kbmi.dta" dataset with the bank ticker as the key variable. After merging, all unmatched observations as well as the leftover "_merge" variable are dropped. Then the bank ticker is encoded into the numerical variable "bank_id", which allows it to be used in declaring the dataset's panel format with "bank_id" as the panel variable and quarter as the time variable. The process is shown in lines 57 to 62.

For this stress testing exercise, we also dropped one outlier observation; namely banks with abnormalities in their public report, shown in line 64. For future use, it is best to drop bank observations that become significant outliers in their income statement or balance sheet reporting as this can create noise that underestimates the industry-level capital adequacy ratio (CAR). Cases such as this often appear in banks from KBMI 1 and 2.

Figure 8. Cleaning bank-level data using Stata

```
66      tsfill, full
67      decode bank_id, generate(bank2)
68      replace bank = bank2
69      drop bank2
70      replace id = bank + " IJ Equity" if id == ""
71
72      order bank bank_id quarter *
73      drop if quarter < tq(2010q1)
74
75      foreach i of varlist _all {
76          local lab: var label `i'
77          if strmatch("`lab'", "*(fill=*") == 1 {
78              local newlab = strupper("`i'")
79              la var `i' "`newlab'"
80          }
81      }
82
83      la var dates "Date"
84      la var bank_id "Bank Ticker"
85
86      save "$data\bank.dta", replace
87
```

From the results of the panel setting, there are time gaps in the observations of the dataset, which is caused by the previously dropped duplicate observations. As shown in Figure 8, to remedy this, the command **"tsfill, full"** is used to create new empty observations that fill all the time gaps in the dataset. However, since the command only fills gaps in the panel variable, the next four lines are used to fill in missing observations in the bank ticker and bank security name variables. Afterwards, the bank ticker and quarter variables are moved to the leftmost columns of the dataset. Finally, a loop is used to label the bank-level variables while the panel and time variables are labelled manually, and the dataset is promptly saved.

Figure 9. Adding new rows to include forecasted data (scenarios) using Stata

```
92
93  use "$data\bank.dta", clear
94
95  egen max=max(quarter)
96  format max %tq
97  label variable max "Maximum period"
98  tsappend, add(4)
99  replace max=l.max if max==.
100 gen year=yofd(dofq(quarter))
101 label variable year "Year"
102 xtset bank_id quarter
103
104 gen npl_r=bs_non_perform_loans/bs_tot_loan*100
105 label variable npl_r "NPL ratio"
106
```

After all the needed data have been stored properly, the next step is to set up additional rows for our forecasted bank dataset based on the scenarios; that is, spaces for future quarters. The data in those future quarters will be filled with the banking forecast predictions. Figure 9 shows – starting from line 95 – the command creates a variable called **"max"**, filled with the most recent quarter in the dataset and then set "max" as a time variable in terms of quarters. To add additional rows of quarter periods, the command **"tsappend"** is employed with specifying **"add(4)"** to add

four additional rows. It is important to also put max as the recent actual quarters for those four additional rows. A variable "year" is also created in line 100 to denote the time in terms of year and then proceed to set the "bank_id" data as a panel data with quarter as the time dimension, illustrated in line 102.

Figure 10. Creating financial ratios using Stata

```
111  gen prov_r=is_prov_for_loan_loss/bs_tot_loan*100
112  la var prov_r "Provision to loans"
113
114  gen c_loan_r=bs_tot_loan/bs_tot_asset*100
115  gen c_dep_r=bs_customer_deposits/bs_tot_asset*100
116  gen c_nii_r=non_int_inc/bs_tot_asset*100
117  gen c_nie_r=non_int_exp/bs_tot_asset*100
118  foreach j in loan_r dep_r nii_r nie_r{
119      egen `j'=mean(c_`j'), by(bank_id)
120      drop c_`j'
121  }
122
123  gen c_lending_r=is_int_inc/bs_tot_loan*100
124  egen lending_r=mean(c_lending_r), by(bank_id)
125  label variable lending_r "Lending rate"
126  gen c_saving_r=(is_int_inc-net_int_inc)/bs_customer_deposits*100
127  replace c_saving_r=0 if c_saving_r<0
128  egen saving_r=mean(c_saving_r), by(bank_id)
129  label variable saving_r "Saving rate"
130  foreach j in lending_r saving_r{
131      replace `j'=. if quarter>max
132      drop c_`j'
133  }
134
135  gen afs_r=av_sa_fa_at_fv_thru_ot_comp_inc/bs_tot_loan*100
136  la var afs_r "AFS securities to assets"
137  gen tra_r=trading_secs_fa_at_fv_thru_pl/bs_tot_loan*100
138  la var tra_r "Trading securities to assets"
139  egen sec_r=rowtotal(afs_r tra_r), missing
140  replace sec_r=100 if sec_r>100
141  la var sec_r "Tradeable securities to assets"
142
143  gen tax_rate=0.22 // TAX RATE ASSUMPTIONS
144  la var tax_rate "Tax rate"
145  save "$data\bank.dta", replace
146
```

The creation of the bank ratio variables and tax assumption can now start, which is shown in Figure 10. The banking indicators for credit risk that will be created are non-performing loans (NPLs), provision to loan, loan to asset, deposit to asset, non-interest income to asset, non-interest expense to asset, lending rate, and savings rate. For market risk the indicators that will be constructed are AFS securities to assets, trading securities to assets, and tradeable securities to assets. Several stages of looping commands are constructed for the credit risk measures, whereby the loops function to obtain the average level of each of those indicators for each bank. After this, the tax ratio assumptions need to be set up. Note that the assumed tax ratio is at 22% of earnings before tax. If there is a need in the future to change the assumptions or there is a change in tax policy, then line 143 should be modified. The last step in this stage is to save the change into "bank.dta".

Credit and asset price risk model

For the modelling of credit risks, this guideline provides two forecasting models: 1) the prediction of NPL; and 2) the prediction of loan loss provision (LLP), which is also known as *cadangan kerugian pengurangan nilai* or CKPN. Before going further with the Stata procedure, it is important

to examine the regression model used for each forecast. The regression equation for the NPL prediction can be observed below:

$$NPLr_{i,t} = \beta_0 + \vartheta_i Bank_i + \beta_1 GDP_{i,t-1} + \beta_2 GDP_{i,t-2} + \beta_3 GDP_{i,t-3} + \beta_4 GDP_{i,t-4} + \gamma_1 reporate_{i,t-1} + \gamma_2 reporate_{i,t-2} + \gamma_3 reporate_{i,t-3} + \gamma_4 reporate_{i,t-4} + \delta_1 \Delta property_{i,t-1} + \delta_2 \Delta property_{i,t-2} + \delta_3 \Delta property_{i,t-3} + \delta_4 \Delta property_{i,t-4} + \epsilon_{i,t}$$

The forecast of NPL in this stress test has 12 auto-regressors and bank fixed effects. *i* denotes bank observation and *t* denotes time observation. There are four regressor variables: GDP growth (GDP in the equation), BI rate (i.e. repo rate), and changes in property prices (i.e. delta property). Each of these regressors will also include their lags, from t-1 until t-4 (four quarters). There are also bank fixed effects (bank-specific characteristics), represented by intercepts depending on the specific bank: the $Bank_i$ is a vector of a dummy variable where it has the value 1 for the specific bank regressed and 0 otherwise. The technical implementation of the regression can be observed in line 207 in Figure 12. The remaining code lines are further explained in the next section. The regression command in Stata starts with the **"qui xi: xtreg"** command, which puts the panel data regression in quiet form, such that the regression result is not posted in the Stata output screen (as we need only the coefficient result for the prediction). The additional option of **"vce()"** is to create robust heteroscedastic standard errors to avoid issues in homoscedasticity that might affect the standard error. The **"boostrap, rep (10) seed (1234)"** options within "vce()" is to specify for bootrap estimation with replications of 10 times and random-number seed so the estimation is consistent when being reproduced.

Figure 11. NPL modelling and forecasting using Stata

```
196     *NPL modelling and forecasting*
197     qui xi: xtreg npl_r l.gdp l2.gdp l3.gdp l4.gdp l.reporate l2.reporate l3.reporate l4.reporate l.d_property l2.d_property l3.d_property l4.d_property,
        fe vce(bootstrap, rep(10) seed(1234))
198     predict npl_b`i', xbu
199     predict npl_i`i', xb
200     gen d_npl_i`i'=d.npl_i`i'
201     replace npl_b`i'=l.npl_b`i'+d_npl_i`i' if npl_b`i'==.
202
```

The next step of this credit risk modelling is to forecast the LLP that the bank has prepared to buffer the risk of poor loans. The regression model is stated in the equation below, and the technical coding is shown in line 207 of Figure 12:

$$Provr_{b,i,t} = \beta_{0,b} + \vartheta_{b,i} Bank_{b,i} + \beta_1 NPLR_{b,i,t} + \beta_2 NPLR_{b,i,t-1} + \beta_3 NPLR_{b,i,t-2} + \beta_4 size_{b,i,t} + \beta_5 reporate_{b,i,t} + \beta_6 reporate_{b,i,t-1} + \epsilon_{b,i,t}$$

The forecast of LLP regression has six auto-regressors and a bank fixed effect. The regressors are the NPL ratio (NPLR) and the lagged version up to t-2, bank size (measured as a log of the assets of the bank), and the BI rate with the lagged t-1. There is an additional indicator *b* taking values of 1,2 and 3, denoting the macroeconomic scenarios that are being employed. As observed below in line 207, the **"b`i'"** represents the scenarios, as the regression will be conducted three times, separately for each scenario, which yields an LLP provision ratio for each bank at each scenario and at each quarter. This process will be conducted in a loop mechanism within Stata that will be explained in the next section. One of the key takeaways is that the bank's LLP is determined by both the NPL and BI rate. Thus, the previous NPL modelling creates future predicted NPL values that determine the predicted LLP allocated by banks. So do the three BI rate scenarios.

Figure 12. Loan loss provision modelling and forecasting using Stata

```
206     *Provision modelling and forecasting*
207     qui xi: xtreg prov_r l2.npl_b`i' l.npl_b`i' npl_b`i' size_b`i' l.reporate_b`i' reporate_b`i', fe vce(bootstrap, rep(10) seed(1234))
208     predict prov_b`i', xbu
209     label variable prov_b`i' "Bank provisions scenario `i'"
210     predict prov_i`i', xb
211     gen d_prov_i`i'=d.prov_i`i'
212     replace prov_b`i'=l.prov_b`i'+d_prov_i`i' if quarter>max
213
```

Modelling for losses arising from tradeable securities is less complex than that of credit risk losses. It boils down to the way tradeable securities risk is tied to changes in bond prices, which for this stress test exercise is a function of the BI rate. As the future values of the BI rates and bond prices have been determined through our macro scenarios, the estimation does not require a regression model. For that purpose, this exercise defines market risk as the formula described in line 215 of Figure 13. The modelled variable "loss" represents the predicted loss from market risk, "d_price" as changes in bond prices, and "sec" as the total available for sale (AFS) and mark to market (MtM) tradeable securities that the bank possesses. This model also includes the looping identifier **"_b`i'"** which ensures the model predicts different market risk losses for every scenario employed. This is the logical effect of varying bond price changes for each scenario, such as what has been constructed in the macro scenario.

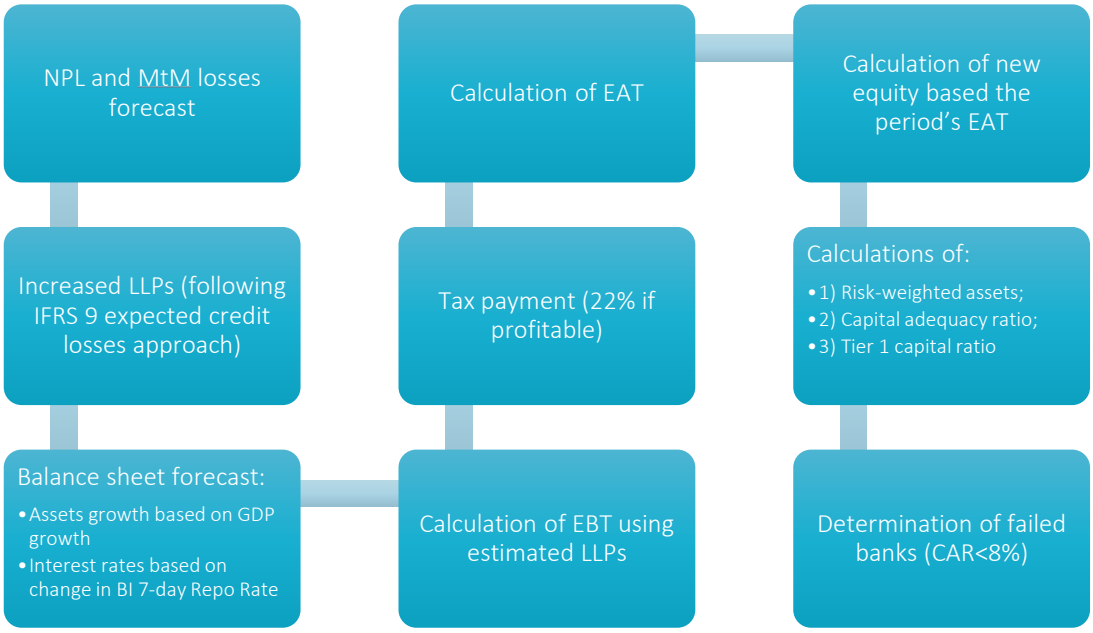
Figure 13. Estimating losses from tradeable securities using Stata

```
214     *Losses from tradeable securities*
215     gen loss_b`i'=d.d_price_b`i'/10000*sec_b`i'
216     label variable loss_b`i' "Losses from securities transactions"
217
```

Balance sheet transmissions

Before taking a deep dive into the technical work, it is vital to properly understand the flows of the stress test transmission, going from the macroeconomic scenario building to losses in banks arising from credit or market phenomena, and then to how these affect the level of capital that banks possess, which in turn influences the banking CAR. The flows of transmission in Figure 14 illustrate how the stress test channels affect a bank's balance sheet. Losses caused by poor credit repayment or tradeable securities push banks to increase their allocated LLP. A continuous increase in LLP will erode the after-tax profit. This profit erosion (or possible overall loss) translates to less capital that a bank may possess in the next year. This reduction (or eroded growth) in the capital level of a bank will continue as the scenario employs a worsening economic condition in each quarter, and eventually a bank will be marked as failed if its CAR level falls under 8%. This flow of transmission indirectly describes the mechanism of how the stress test measures the resiliency of banks. Banks that have sufficient capital to buffer constant falling performance demonstrate their financial capacity in the face of adverse situations. An important caveat is that this scenario does not represent a true possible future. Rather, it is no more than a method to measure the banking sector's resiliency under an arbitrarily constructed stress scenario.

Figure 14. Flows of balance sheet transmission in banking stress test



For a proper understanding of how to implement the transmission using Stata, this section circles back the procedure to line 151, as illustrated in Figure 15. Starting from this line, the macro scenarios will be introduced to affect the profitability and balance sheet of banks in the banking data. This entire stage takes most of the entire stress testing process, including the previously discussed NPLs, provision, and securities loss models. Those models are part of the looping process that starts in line 155 and goes on to line 251. The first step within the start of the loop command is to merge the "bank.dta" with each scenario's data in "comb1.dta", "comb2.dta" and "comb3.dta".

Figure 15. Estimating different scenarios using a looping command in Stata

```
147  ****************************
148  ******** Simulations ********
149  ****************************
150
151  use "$data\bank.dta", clear
152
153  xtset bank_id quarter
154
155  forval i=1/3{
156
157      *Obtaining scenario*
158      merge m:m quarter using "$data\comb`i'.dta", force
159      keep if _merge==3
160      drop _merge
161      foreach j in gdp reporate cpi forex d_property ln_forex d_price{
162          gen `j'_b`i'=`j'
163      }
164      sort bank_id quarter
165      xtset bank_id quarter
```

Now that the scenarios have been integrated, the process goes to the next step, which is to calculate various balance sheet components. Each scenario will generate different ratios and balance sheet components. This process takes place from line 168 to 194 in Figure 16. In this step Stata will calculate the total asset, risk weighted assets, loan to asset, deposit to asset, securities position, non-interest income, and non-interest expenses for each bank and for each of the three scenarios. In a more detailed explanation, differences in GDP growth scenarios will create different asset growth and different changes in the risk-weighted assets – all of these will change the loan to asset and the deposit to asset.

Figure 16. Balance sheet calculations using Stata

```
167     *B/S calculations*
168     gen asset_b`i'=bs_tot_asset
169     bys bank_id (quarter): replace asset_b`i'=asset_b`i'[_n-1] * (1+gdp/100) if missing(asset_b`i') & quarter>max
170     label variable asset_b`i' "Bank assets scenario `i'"
171
172     gen rwa_b`i'=bs_risk_weighted_assets
173     bys bank_id (quarter): replace rwa_b`i'=rwa_b`i'[_n-1] * (1+gdp_b1/100) if missing(rwa_b`i') & quarter>max
174     label variable rwa_b`i' "RWA scenario `i'"
175
176     gen size_b`i'=log(asset_b`i')
177     label variable size_b`i' "Bank size scenario `i'"
178
179     gen loan_b`i'=loan_r/100*asset_b`i'
180     label variable loan_b`i' "Bank loans scenario `i'"
181
182     gen dep_b`i'=dep_r/100*asset_b`i'
183     label variable dep_b`i' "Bank deposits scenario `i'"
184
185     gen sec_b`i'=sec_r
186     label variable sec_b`i' "Bank securities scenario `i'"
187
188     bys bank_id (quarter): replace sec_b`i'=sec_b`i'[_n-1] if missing(sec_b`i') & quarter>max
189
190     gen nii_b`i'=non_int_inc
191     replace nii_b`i'=l.nii_b`i' if missing(nii_b`i') & quarter>max
192
193     gen nie_b`i'=non_int_exp
194     replace nie_b`i'=l.nie_b`i' if missing(nie_b`i') & quarter>max
```

Figure 17. Income statement and capital adequacy calculations using Stata

```
219     *I/S calculations*
220     gen lending_r_b`i'=lending_r
221     bys bank_id (quarter): replace lending_r_b`i'=lending_r_b`i'[_n-1] + d.reporate if missing(lending_r_b`i') & quarter>max
222
223
224     gen saving_r_b`i'=saving_r
225     bys bank_id (quarter): replace saving_r_b`i'=saving_r_b`i'[_n-1] + d.reporate if missing(saving_r_b`i') & quarter>max
226
227     gen int_inc_b`i'=lending_r_b`i'/100*loan_b`i'
228     gen int_exp_b`i'=saving_r_b`i'/100*dep_b`i'
229
230     gen nim_b`i'=(int_inc_b`i'-int_exp_b`i')/asset_b`i'
231
232     gen ebt_b`i'=int_inc_b`i' - int_exp_b`i'+ nii_b`i' - nie_b`i' - (prov_b`i'/100)*asset_b`i' - loss_b`i'
233     gen tax_b`i'=ebt_b`i'*tax_rate
234     replace tax_b`i'=0 if tax_b`i'<0
235     gen eat_b`i'=ebt_b`i'-tax_b`i'
236
237     *CAR calculations*
238     gen tier1_b`i'=bs_tier1_capital
239     replace tier1_b`i'=l.tier1_b`i'+eat_b`i' if quarter>max
240     label variable tier1_b`i' "Tier 1 capital scenario `i'"
241     gen cap_b`i'=bs_tot_cap_fund
242     replace cap_b`i'=l.cap_b`i'+eat_b`i' if quarter>max
243     label variable cap_b`i' "Regulatory capital scenario `i'"
244
245     gen t1r_b`i'=tier1_b`i'/rwa_b`i'*100
246     label variable t1r_b`i' "Tier 1 ratio scenario `i'"
247     gen car_b`i'=cap_b`i'/rwa_b`i'*100
248     label variable car_b`i' "CAR scenario `i'"
249
250     drop d_npl_i`i' prov_i`i' d_prov_i`i' gdp reporate cpi forex property d_property ln_forex d_price
251 }
252
```

Continuing this process, various income statement components will also be calculated according to the implemented scenarios. The three different scenarios will yield three different levels of credit losses and market losses which create three scenarios of bank earnings after tax. As can be seen in the do file below, from line 220 to 235 in Figure 17, our previous risk modelling plays a significant role in this calculation of earnings. The previously modelled provision will predict three scenarios of bank provision which yields three scenarios of bank earnings. The process then goes to the last and final procedure: the capital components calculations. The three earnings scenarios will logically create three scenarios of capital levels, tier 1 capital ratios, and the CARs. Thus, the entire technical procedure of the banking solvency stress test has concluded with three CAR scenarios for each bank. Whichever bank has its CAR level fall below 8% in any of these scenarios (in the future forecasted quarters) will be considered to have failed. The steps to get to the capital calculation

are illustrated in lines 238 to 251 in Figure 17. Users should then proceed to interpret the results and illustrate them graphically, which will be explained in the following section.

5. Results and interpretations

The framework outlined in this report examines the impact of various macroeconomic shocks on the banking sector. Since the framework utilises bank-level data, the analysis can be extended to KBMI level or even to that of the individual banks included in the sample/exercise. Nevertheless, considering the focus of the Secretariat on financial stability, the analysis of the results should start with the industry-wide effects.

Figure 18 illustrates the results using 2024 Q2 data. Stress test results therefore start from 2024 Q3 until 2025 Q2 (four subsequent quarters from the last observed data). Panel A of Figure 18 depicts the evolution of NPLs during stress. Under Scenario 1, each bank's NPLs range between 0.17% and 8.5%, while the industry average fluctuates around 3%. Under Scenario 2, the industry average increases to around 3.2%, while some banks can have NPLs up to 8.7%. Under Scenario 3, each bank's NPLs can reach 9.1%, which is almost two-fold higher than the indicative threshold of 5%. At the industry level the ratio reaches 3.4%. Panel B illustrates total gains and losses of MtM securities. As shown in the previous section, the calculation of these market losses from asset prices follows the IFRS 9 fair value through profit or loss (FVTPL) and fair value through other comprehensive income (FVOCI) approaches, while the results are presented as the percentage of total assets.

The sum of the losses from credit and asset prices risks is presented in Panel C of Figure 18. These charts not only show the percentage of losses with respect to total assets in each scenario, but also the contribution of each (credit or asset price) risk to total losses. Losses from credit risk are derived from the additional provisions (CKPN) that need to be acknowledged by banks in response to higher NPLs. Overall, total losses fluctuate between 0.4% and 0.5% of total assets, while market losses from asset prices can reach up to 15% of total losses. This estimate suggests that credit risk still dominates the banking sector's overall risk.

After the calculation of total losses, the analysis shifts to the effect of these losses on bank profitability (net income). The estimates are shown in Panel D of Figure 18. Overall, the results show that Scenario 1 has the mildest impact on profitability, while Scenario 3 has the most severe effect. These effects align with the magnitude of each of the estimated additional provisions and losses from MtM securities under each scenario.

Finally, Panels D and E illustrate the dynamic effects of all scenarios on CAR and tier 1 capital ratio respectively. The results show that even under the worst scenario, both capital ratios are still above 22%, reflecting the resilience and sufficiency of the capital absorption buffers of Indonesian banks. Table 4 provides the summary statistics for the stress test results.

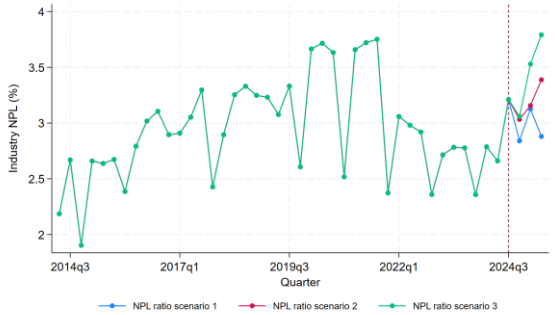
Table 4. Stress test results

Scenario	NPL (%)	Asset prices losses (%)	CAR (%)
2024Q2	2.79	0.00	39.15
Scenario 1	3.02	−0.02	31.72
Scenario 2	3.20	−0.01	31.66
Scenario 3	3.40	0.01	31.60

Figure 18. Industry-wide stress test results

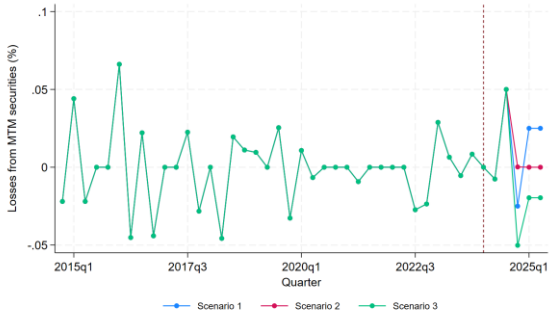
A) Non-performing loans

% of total assets



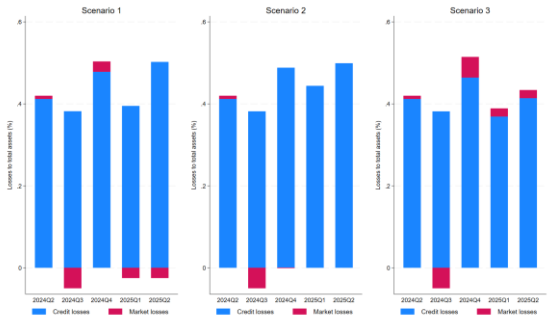
B) Gains/losses of mark-to-market securities

% of total assets



C) Total costs of credit and market risks

% of total assets



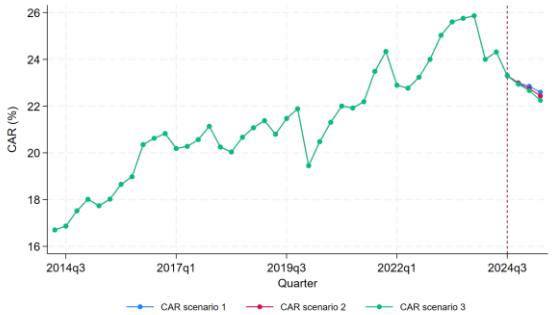
D) Net income

% of total assets



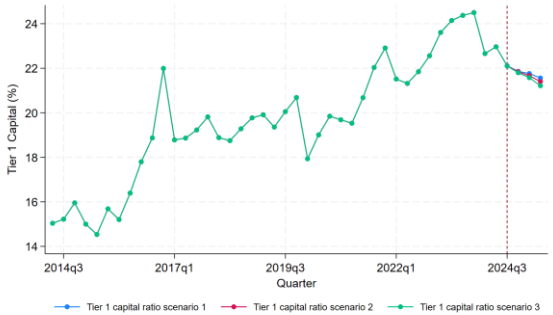
E) Capital adequacy

% of total risk-weighted assets



F) Tier 1 capital

% of total risk-weighted assets



6. Conclusions

The Global Financial Crisis of 2007–2009 prompted the development of better stress testing as part of prudential supervision and regulation. Since then, policymakers and academic experts have made substantial efforts to improve all aspects of stress testing design.

This report has presented the important aspects of stress testing from a conceptual perspective. It then provided both conceptual and technical guidelines for designing a simplified yet credible stress testing framework using publicly available data. These guidelines include scenario building, credit risk modelling, data collection, and results interpretation. Note that data limitations can be a major constraint to designing a good stress testing framework. When this is the case, appropriate modelling and careful interpretation can still yield credible results useful for policymakers.

Finally, even though this report points out some key underpinning features of stress testing, there is no one-size-fits-all design for stress testing. Stress testing can show substantial variations depending on the individual characteristics of a country's financial system. Therefore, some degree of flexibility is necessary to design stress testing that fits with the financial system in question.

After more than a decade of implementation, bank stress testing remains an important prudential tool. However, the credibility of stress testing largely depends on whether prudential authorities can create sensible macroeconomic scenarios, provide accurate and granular data, design robust econometric models, and ensure the testing results are appropriately used to meet the stated objectives.

Future developments

The stress testing framework documented in this report has addressed two important aspects of bank stress testing: credit risk and security prices volatility. In practice, however, stress testing also covers other risks such as currency mismatch, interest rate movements, and maturity mismatches. Addressing these risks requires more granular data that is not available publicly. For this reason, the acquisition of more granular data from supervisors is required. The availability of such data would not only enable the calculation of additional bank risks, but also improve the modelling of existing risks (such as credit risk and security prices volatility).

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Acknowledgements

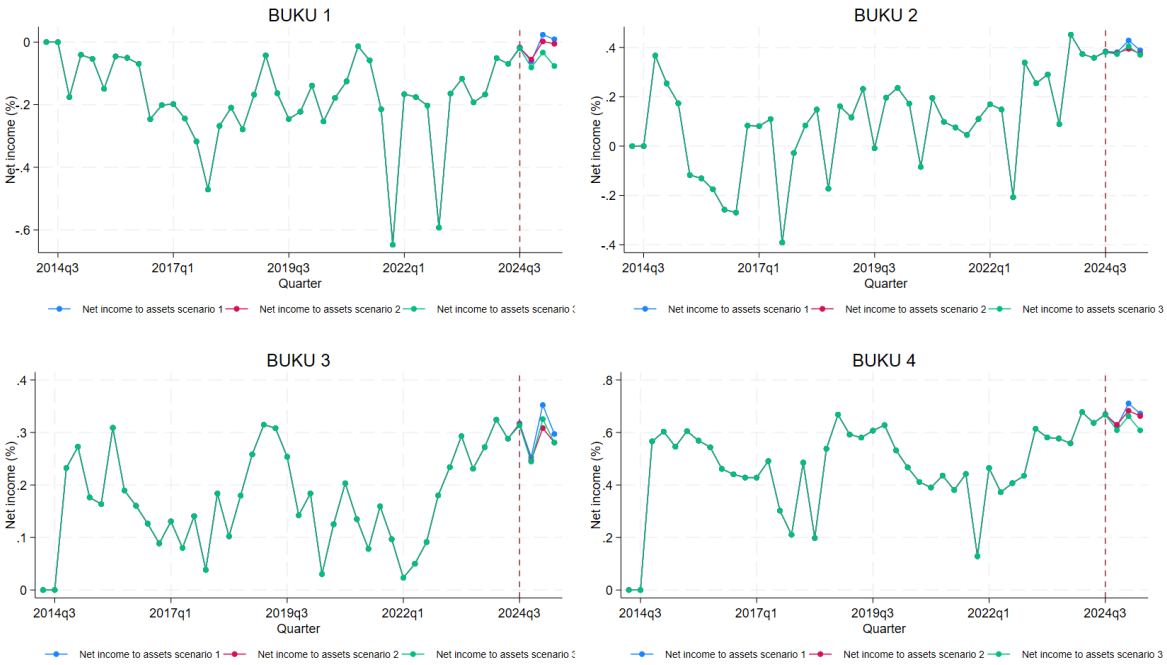
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Appendix

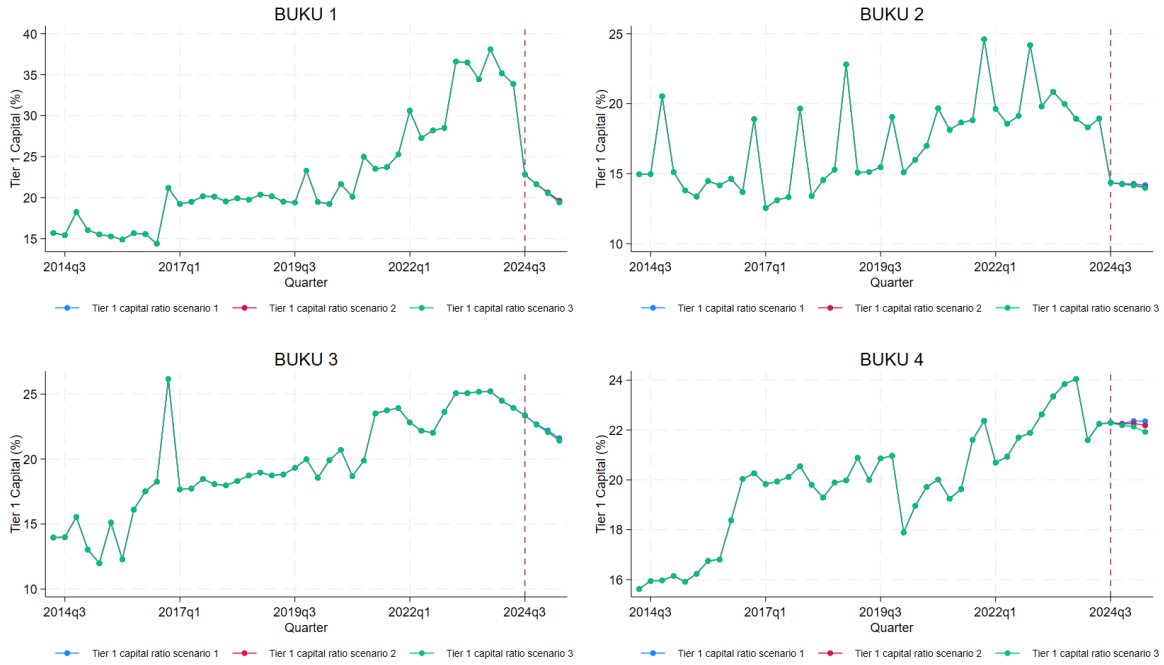
Appendix 1. NPL ratio by KBMI



Appendix 2. Net income to assets by KBMI



Appendix 3. Tier 1 capital ratio by KBMI



Appendix 4. Capital adequacy ratio by KBMI

